

SYNOPSIS

The UK registered, 1882 grt, general cargo vessel *Scot Explorer* sailed from Corpach, Scotland on 30 October 2004 for passage to Varberg, Sweden, carrying 2107 tonnes of timber logs. At 1840(UTC+1) on 2 November 2004, she collided with the 14.7m Danish fishing vessel *Dorthe Dalsoe* in international waters in the Kattegat. *Dorthe Dalsoe* sustained considerable damage to her bow and required assistance from a Danish patrol vessel and other vessels in the vicinity, being subsequently towed safely into port. The damage to *Scot Explorer* was superficial and she was able to continue her voyage.

The collision occurred when *Scot Explorer* was following 'Route T' on a course of 131° at 9.8 knots, and *Dorthe Dalsoe* was on a course of 260° in autopilot at 7 knots. The fishing vessel was returning to Denmark to land her catch of prawns, after being at sea since 0300(UTC+1) the previous day. It was dark and visibility was good.

At the time of the accident, the crew of *Dorthe Dalsoe*, which comprised her skipper and an inexperienced deckhand, were working on the vessel's shelter deck, from where they were unable to see any vessels forward of the beam. Before leaving the wheelhouse, her skipper had seen a masthead light on the starboard bow but did not take any further action to determine if a risk of collision existed. He also configured the navigation lights to indicate that the vessel was not under command on the assumption that other vessels would keep out of the way.

The master of *Scot Explorer*, who was on watch on the bridge, had detected *Dorthe Dalsoe* both visually and by radar. He had assessed that she would pass between 3 and 4 cables down the port side. When the fishing vessel had closed to within 1 mile, the master attended to some routine work at the chart table. When he next looked out of the window, about 2 minutes later, he saw *Dorthe Dalsoe* very close off the port bow. Although the master of *Scot Explorer* immediately changed from auto to manual steering and applied starboard helm, collision could not be avoided.

The investigation highlighted several causal and contributory factors. These included:

Dorthe Dalsoe

- Her skipper might have been fatigued as a result of insufficient and poor quality sleep during the period leading up to the accident.
- Her skipper's knowledge of the COLREGS with regard to lookout, the use of radar, and the use of 'not under command' lights was poor, and its application dangerous.
- The navigation lights displayed were confusing.
- Her starboard side navigation light was probably not discernible among her bright white deck lights.
- A proper lookout was not maintained and a risk of collision could not be determined by the skipper when working on the shelter deck.

Scot Explorer

- The estimation of the CPA of *Dorthe Dalsøe* by radar was inaccurate because of the rudimentary methods used. Had more of the radar facilities available been used, a more accurate assessment of the CPA, and therefore risk of collision, would have been possible.
- The acceptance of a CPA of 4 cables was inappropriate given the sea room available and the lack of other shipping in the vicinity.
- The master of *Scot Explorer* was distracted as the vessels closed.
- The master was alone on the bridge. Had an additional lookout been on the bridge, he would have been well placed to alert the master to the approach of the fishing vessel.
- The AB nominated as lookout on the bridge of *Scot Explorer* was unavailable because priority was given to his duties in the galley.
- Although manned in accordance with her safe manning document, it was difficult for the vessel to comply with the STCW 95 requirements for a dedicated bridge lookout.

Following the collision, the ship managers of *Scot Explorer* increased the manning of the vessel above the minimum required by her safe manning document, and issued guidance on the use of bridge lookouts, and the determination of safe passing distances. The Danish Maritime Authority stated its intention to promulgate the lessons learned from this and other accidents involving Danish fishing vessels, to the Danish fishing industry, on completion of a safety study currently in progress.

In response to recommendations previously made by the MAIB, the Maritime and Coastguard Agency has commissioned an independent survey of safe manning levels across Europe. Depending on the outcome of this study, a report will be forwarded to the IMO highlighting the inconsistencies identified. The Maritime and Coastguard Agency also intends to issue guidance on the use of lookouts in an MGN.

In view of the actions already taken, a recommendation has been made to the MCA to maintain the priority given to those actions currently in progress regarding safe manning and the use of bridge lookouts. A recommendation has also been made to the International Chamber of Shipping to promulgate to its members the need to ensure that ships are able to meet the requirements for a dedicated lookout during darkness, and that bridge equipment must be used to its full potential if passing distances are to be accurately determined.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF SHIPS AND ACCIDENT

***Scot Explorer* (Figure 1)**

Vessel details

Registered owner	:	Scot Explorer Shipping Ltd
Manager	:	Intrada Ships Management Ltd
Port of registry	:	Rochester
Flag	:	UK
Type	:	General cargo
Built	:	1996, Societatea Comerciala Navol S.A.
Classification society	:	Germanischer Lloyd
Construction	:	Steel
Length overall	:	81.68m
Gross tonnage	:	1882
Engine power	:	749kW

***Dorthe Dalsoe* (Figure 2)**

Vessel details

Registered owner	:	Privately owned
Port of registry	:	Frederikshavn
Flag	:	Denmark
Type	:	Side trawler
Built	:	1962
Construction	:	Wood
Length overall	:	14.7m
Gross tonnage	:	19.7
Engine power and/or type	:	125kW

Figure 2



Dorthe Dalsøe

Accident details

Time and date	:	1840(UTC+1) on 2 November 2004
Location of incident	:	57°30.9N 011°17.0E
Persons on board	:	<i>Scot Explorer</i> – 5 <i>Dorthe Dalsøe</i> – 2
Injuries/fatalities	:	Nil
Damage	:	<i>Scot Explorer</i> – Superficial <i>Dorthe Dalsøe</i> – The stem post was broken and the fore deck crushed. Damage was most extensive on the port side of the bow where the hull was breached close to the waterline.

1.2 NARRATIVE

(All times are UTC+1, and all courses are true)

1.2.1 *Scot Explorer*

Scot Explorer sailed from Corpach, Scotland on 30 October 2004, for passage to Varburg, Sweden. She was carrying 2107 tonnes of timber logs, of which 226 tonnes was secured on deck. During the evening of 2 November, the ship was transiting Route 'T' in the Kattegat on a course of 131° in autopilot; speed over the ground was 9.8 knots (**Figure 3**). The chief officer was the OOW. At 1820, he saw a vessel at a long distance on his port bow. The vessel, which was closing, was brightly lit by her white deck lights. A red light was visible above these lights.

The chief officer assessed that the vessel was a fishing vessel, but could not determine her aspect or movement, either visually or by radar. As a precaution, he switched on *Scot Explorer*'s deck lights to help ensure that she could be clearly seen by the approaching vessel. Shortly after, when the fishing vessel was between 3 and 4 miles, the chief officer saw a second red light on the fishing vessel. He assumed this to be her port side navigation light. Based on the sighting of this light, together with inspection of the fishing vessel's relative trail on the radar display, the chief officer assessed that the fishing vessel had altered course to starboard, and would pass at a distance of 3 to 4 cables down the port side.

When the fishing vessel was at a range of between 2.5 and 3 miles, the master arrived on the bridge to take the bridge watch. The chief officer showed him the 1800 position on the paper chart, which was the last plotted position, and also pointed out the fishing vessel, two points off the port bow. The chief officer then went to the mess room for his evening meal.

The master assessed the fishing vessel to be on a reciprocal course and passing between 3 and 4 cables down the port side. He did not think that the vessel was actually engaged in fishing because of her speed, which he estimated to be about 7 knots. There were no other vessels in close proximity. When the fishing vessel was at one mile, the master could see a red light at the top of her masthead, a white masthead light, what he took to be a red sidelight, and deck lights. At this point the master went to the chart table, sited on the port side of the bridge, to monitor the ship's position using the graphical display on a GPS receiver. He then updated the ship's official logbook. After being at the chart table for about 2 minutes, the master looked up and saw the fishing vessel between 30m and 40m off the port bow. He immediately changed to hand steering and applied starboard helm. Seconds later the vessels collided.

1.2.2 *Dorthe Dalsøe*

Dorthe Dalsøe sailed from Frederikshavn, Denmark at 0300 on 1 November for passage to the fishing grounds in the southern part of the Vinga Rende, where she arrived at 0700 the same morning. She then trawled for prawns, with each tow lasting for about 7 hours. After hauling the fishing gear at 1800 on 2 November, the vessel commenced her return passage to Frederikshavn. Her course was 260° in autopilot, and her speed was 7 knots. There were no radar contacts on either of the two radar displays fitted, which were set to 1.5 and 3 mile range scales. However, the skipper saw a white masthead light about 45° on the starboard bow and at an estimated range of between 4 and 5 miles. The skipper switched off the vessel's green trawling light, and switched on two red lights sited one above the other on the aft masthead. He then joined the deckhand on the shelter deck (**Figure 4**). The next he knew was when both crew felt a heavy bump as *Dorthe Dalsøe* and *Scot Explorer* collided.

In addition to the two red lights on the aft masthead, the vessel displayed a white light on her foremast, red and green sidelights, a white stern light, and deck lights. The skipper frequently displayed this configuration of lights to indicate that the vessel was 'not under command' when he left the wheelhouse unattended to clean the catch on the shelter deck. It was the skipper's expectation that other vessels would keep out of the way. There were no reported problems with the functioning of the vessel's autopilot.

Figure 4



The shelter deck on board *Dorthe Dalsøe*

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE

As the skipper of *Dorthe Dalsoe* had only 8 hours sleep, taken in the wheelhouse when towing, during the 38 hours before the accident, it is probable that he was feeling some effects of fatigue. The quantity of sleep was insufficient, and its quality was poor. However, in view of the fact that it was his normal practice to leave the wheelhouse unattended when cleaning the catch, it is difficult to determine the extent to which the effects of fatigue might have influenced the skipper's decision-making in this respect.

Assuming that the recorded hours of rest (**Annex A**) for the master of *Scot Explorer* are accurate, these exceed the minimum required by STCW 95 and MGN 1767 (M). Additionally, as the master had also slept for 2.5 hours in the afternoon, did not feel tired, and was active on the bridge after taking over the watch, there is no evidence to indicate that his performance was affected by fatigue.

2.3 INTERPRETATION OF THE SITUATION

Figure 8 is a plot based on the stated courses and speeds of *Scot Explorer* and *Dorthe Dalsoe*. This shows that *Dorthe Dalsoe* closed *Scot Explorer* from a bearing of 110° with a relative closing speed of 15.2 knots. This correlates with the assessment that she was two points on the port bow when the master of *Scot Explorer* took over the watch.

The course and speed of *Scot Explorer* have been verified by AIS data (**Figure 5**). Similar data is not available for *Dorthe Dalsoe*, and it is possible that the fishing vessel altered course when the vessels were in close proximity due to a malfunction of her autopilot. However, as the skipper of *Dorthe Dalsoe* was not in a position to detect or notice such a malfunction, and the master of *Scot Explorer* was not observing the vessel at the time, there is no eye witness evidence to corroborate this possibility. Also, given that there had been no previous problems experienced with the autopilot on board *Dorthe Dalsoe*, its malfunction is considered to have been unlikely.

It is therefore most probable that both vessels had been on steady courses throughout the build up to the collision.

2.4 WATCHKEEPING PRACTICES ON BOARD *DORTHE DALSOE*

A vessel “not under command” is defined in the COLREGS (Rule 3f), and results from an “exceptional circumstance” which prevents a vessel from being able to manoeuvre to avoid other ships. This is normally interpreted as unforeseen and unplanned events such as machinery breakdowns. The cleaning of a fishing vessel’s catch is neither unforeseen, nor unplanned, nor does it affect manoeuvrability. *Dorthe Dalsoe* was therefore not a vessel “not under command” and, as she was also not “engaged in fishing” (Rule 3 d), she was a “power driven vessel” (Rule 3b). As such, given the disposition and movement of the two vessels, she was required to keep clear of *Scot Explorer* in accordance with rules 15 and 16 of the collision regulations (**Annex E**).

The lights displayed by *Dorthe Dalsoe* to indicate that she was “not under command” were also not in accordance with the requirements of the COLREGS (Rule 27a). The display of a white masthead light forward was additional to the requirements, and was a potential source of confusion to other vessels. It was therefore not surprising that both the chief officer, and the master of *Scot Explorer*, were not only unable to identify the lights exhibited as being for a vessel “not under command”, but also misidentified one of the red lights on the aft mast as a sidelight.

When the skipper left the wheelhouse and went to the shelter deck, from where he had virtually no external visibility, he could not maintain a proper lookout, or determine if a risk of collision existed. Although the COLREGS require these obligations to be met by every vessel, regardless of status or condition, such obligations were either not understood by the skipper, or were ignored by him. This was emphasised by the fact that the skipper went to the shelter deck even though he was aware that *Dorthe Dalsoe* was approaching Route “T” and the masthead light of *Scot Explorer* was seen closing on the starboard bow. Had the skipper increased the range set on his radar displays as required by COLREG Rule 7, this would have provided him with the information required to determine the CPA of *Scot Explorer*, and the risk of collision.

It is evident that the skipper’s knowledge and application of the COLREGS was poor. His categorisation of *Dorthe Dalsoe* as being “not under command” was erroneous, but was probably seen as a convenient measure, which the skipper thought justified the abrogation of his duties towards other vessels while preparing his catch for landing. The skipper’s decisions to move to the shelter deck, and to display what he considered to be lights indicating that the vessel was not under command, in the expectation that other vessels would keep out of his way, were significant contributory factors to this collision.

collision. However, if the master was only at the chart table for 2 minutes before the collision occurred, it follows that *Dorthe Dalsoe* must have initially been at a distance of only 5 cables. It is possible that the master was at the chart table for a longer period than he recollects, but it is equally possible that he reduced to the 1.5 mile range scale, rather than the 3 mile range scale. If this were the case, the use of the fixed range rings to measure distance would have caused the master to double not only the distance of *Dorthe Dalsoe*, but also her estimated CPA.

Had more of the radar's facilities, such as automatic plotting, electronic bearing line, and variable range marker been used, a more accurate assessment of the CPA of *Dorthe Dalsoe*, and therefore risk of collision, would have been possible. The monitoring of the fishing vessel's bearing by radar would have also given a reliable indication in this respect.

2.5.3 Acceptance of CPA

An estimated passing distance of between 3 to 4 cables does not provide much of a safety margin considering the accuracy of the methods available to determine CPA, and the possibility of unpredictable movement due to mechanical failure or human behaviour. Notwithstanding the master and chief officer's incorrect determination of the heading and CPA of *Dorthe Dalsoe*, given that there were no other vessels in the vicinity, there was no reason why the chief officer or master could not have taken early action to provide a greater margin of safety. Had they done so, the risk of collision would have been considerably reduced.

2.5.4 Lookout

When two vessels are in close proximity, it is usual for an OOW to monitor the situation carefully, at least until the other vessel has passed her CPA. In this case, the master's preoccupation with routine administration distracted him from this important task. He did not monitor *Dorthe Dalsoe* from the time when he moved to the chart table until seconds before the collision. Consequently, because he was alone on the bridge, and had not utilised the radar guard zone facility, he was ignorant of the approaching danger. Had a proper lookout been maintained, the likelihood of collision would probably have been detected in time for successful avoiding action to be taken.

2.6 EMPLOYMENT OF RATINGS AS LOOKOUT

Contrary to the requirements of STCW 95 and the Intrada Ship Management Fleet Circular issued in June 2004, which require an additional lookout to be on the bridge during the hours of darkness, the chief officer and master kept their watches alone during the evening of 2 November. With *Scot Explorer* operating in northern Europe in winter with only two ABs available, the difficulty experienced in providing an additional lookout during the extended hours of darkness was understandable, particularly during meal times. However, there

SECTION 3 - CONCLUSIONS

3.1 FINDINGS

1. Although the quantity of sleep achieved by the skipper of *Dorthe Dalsoe* was insufficient, and its quality was poor, it is difficult to determine the extent to which the effects of fatigue might have influenced his decision-making. [2.2]
2. There is no evidence to indicate that the performance of the master of *Scot Explorer* was affected by fatigue. [2.2]
3. Given the courses and speeds of the two vessels, *Dorthe Dalsoe* closed *Scot Explorer* from a bearing of 110° with a relative closing speed of 15.2 knots. There is no evidence to indicate that *Dorthe Dalsoe* changed course when the vessels were in close proximity due to malfunction of her autopilot. [2.3]
4. *Dorthe Dalsoe* was a 'power driven vessel' not 'a vessel not under command' and, given the disposition and movement of the two vessels, she was required to keep clear of *Scot Explorer* in accordance with rules 15 and 16 of the collision regulations. [2.4]
5. The lights displayed by *Dorthe Dalsoe* to indicate that she was "not under command" were not in accordance with collision regulations, and the display of a white masthead light forward was a potential source of confusion to other vessels. [2.4]
6. When the skipper left the wheelhouse and went to the shelter deck from where he had virtually no external visibility, he could not maintain a proper lookout, or determine if a risk of collision existed. [2.4]
7. The decisions of the skipper of *Dorthe Dalsoe* to move to the shelter deck, and to display what he considered to be lights indicating that the vessel was not under command, in the expectation that other vessels would keep out of his way, were significant contributory factors to this collision. [2.4]
8. The assessments of the master and chief officer of *Scot Explorer* with regard to the CPA and risk of collision with *Dorthe Dalsoe*, which were based on their interpretation of the navigation lights seen, and from a visual inspection of the synthetic trail behind her echo on the radar display, were incorrect. [2.5.1]
9. It is probable that the fishing vessel's green starboard light was not seen by the crew of *Scot Explorer* because it was not readily discernible among the bright white deck lights. [2.5.1]
10. Many fishing vessels display lights and shapes to indicate that they are engaged in fishing when on passage, and it is not surprising that many seafarers are becoming increasingly sceptical of their validity. [2.5.1]

11. Had more of the radar's facilities on board *Scot Explorer*, such as automatic plotting, electronic bearing line, and variable range marker been used, a more accurate assessment of the CPA of *Dorthe Dalsoe*, and therefore risk of collision, would have been possible. [2.5.2]
12. The radar guard facility was not being utilised. [2.5.4]
13. Notwithstanding the incorrect determination of the heading and CPA of *Dorthe Dalsoe*, given that there were no other vessels in the vicinity, there was no reason why the chief officer or master of *Scot Explorer* could not have taken early action to provide a greater margin of safety. [2.5.3]
14. Had the master of *Scot Explorer* maintained a proper lookout, the likelihood of collision would probably have been detected in time for successful avoiding action to be taken. [2.5.4]
15. There was no apparent reason why the conflicting duties of the ABs on board *Scot Explorer* could not have been resolved by better management of the meal times on board. The fact that this was not done, possibly reflects a low priority given to the importance of an additional lookout. [2.6]
16. Had the AB been on the bridge of *Scot Explorer* and not in the galley, he would have been available to assist the master in monitoring *Dorthe Dalsoe*, and would have been well placed to quickly alert him to the approach of the fishing vessel. [2.6]
17. Until OOWs understand and appreciate the contributions to ship safety an additional lookout can make, the positive impact of any regulatory requirements for their use, will be considerably reduced. [2.6]
18. In order to fulfil his duties as, bridge lookout during darkness, cook, and general deckhand employed on maintenance and cargo handling, the AB in question on board *Scot Explorer* was not able to achieve the minimum hours of rest required. [2.7]
19. Given the hours required to be worked by the AB employed in the galley of *Scot Explorer*, and the difficulties experienced in providing an additional lookout, the doubts of the ship manager regarding the ship's manning level were justified. [2.7]
20. The failure of Administrations to check the downward movement of manning levels has resulted in 'safe manning' levels becoming 'unsafe'. [2.7]
21. The detrimental effects on safety, of the downward movement of manning levels, can only be relieved via prescriptive requirements regarding the number of officers and ratings to be carried. Such prescription needs to be applied by all Administrations, or to all ships operating within a geo-economic area such as the European Union. [2.7]